

the preceding figure, occurs in the upper left corner as the subdiagram with labels 1 through 4. This subdiagram has the symmetry of Poincaré duality as well.

From the diagram one can with some effort work out the cup product structure in $H^*(SO(7); \mathbb{Z})$, but the answer is rather complicated, just as the diagram is:

$$F[x, y, z, v, w] / (2x, 2v, x^4, y^4, z^2, v^2, w^2, xz, vz, vw, y^2w, x^3y^2v, \\ y^2z - x^3v, xw - y^2v - x^3v)$$

where x, y, z, v, w have dimensions 2, 3, 7, 7, 11, respectively. It is curious that the relation $x^3 = y^2$ in $H^*(SO(5); \mathbb{Z})$ no longer holds in $H^*(SO(7); \mathbb{Z})$.

Exercises

1. Show that $H^*(K(\mathbb{Z}_m, 1); \mathbb{Z}_k)$ is isomorphic as a ring to $H^*(K(\mathbb{Z}_m, 1); \mathbb{Z}_m) \otimes \mathbb{Z}_k$ if k divides m . In particular, if m/k is even, this is $\Lambda_{\mathbb{Z}_k}[x] \otimes \mathbb{Z}_k[y]$.
2. In this problem we will derive one half of the classification of lens spaces up to homotopy equivalence, by showing that if $L_m(\ell_1, \dots, \ell_n) \simeq L_m(\ell'_1, \dots, \ell'_n)$ then $\ell_1 \cdots \ell_n \equiv \pm \ell'_1 \cdots \ell'_n k^n \pmod{m}$ for some integer k . The converse is Exercise 29 for §4.2.
 - (a) Let $L = L_m(\ell_1, \dots, \ell_n)$ and let $\mathbb{Z}_m^*$ be the multiplicative group of invertible elements of $\mathbb{Z}_m$. Define $t \in \mathbb{Z}_m^*$ by the equation $xy^{n-1} = tz$ where x is a generator of $H^1(L; \mathbb{Z}_m)$, $y = \beta(x)$, and $z \in H^{2n-1}(L; \mathbb{Z}_m)$ is the image of a generator of $H^{2n-1}(L; \mathbb{Z})$. Show that the image $\tau(L)$ of t in the quotient group $\mathbb{Z}_m^* / \pm (\mathbb{Z}_m^*)^n$ depends only on the homotopy type of L .
 - (b) Given nonzero integers $k_1, \dots, k_n$, define a map $\tilde{f}: S^{2n-1} \rightarrow S^{2n-1}$ sending the unit vector $(r_1 e^{i\theta_1}, \dots, r_n e^{i\theta_n})$ in $\mathbb{C}^n$ to $(r_1 e^{ik_1\theta_1}, \dots, r_n e^{ik_n\theta_n})$. Show:
 - (i) $\tilde{f}$ has degree $k_1 \cdots k_n$.
 - (ii) $\tilde{f}$ induces a quotient map $f: L \rightarrow L'$ for $L' = L_m(\ell'_1, \dots, \ell'_n)$ provided that $k_j \ell_j \equiv \ell'_j \pmod{m}$ for each j .
 - (iii) f induces an isomorphism on π_1 , hence on $H^1(-; \mathbb{Z}_m)$.
 - (iv) f has degree $k_1 \cdots k_n$, i.e., f_* is multiplication by $k_1 \cdots k_n$ on $H_{2n-1}(-; \mathbb{Z})$.
 - (c) Using the f in (b), show that $\tau(L) = k_1 \cdots k_n \tau(L')$.
 - (d) Deduce that if $L_m(\ell_1, \dots, \ell_n) \simeq L_m(\ell'_1, \dots, \ell'_n)$, then $\ell_1 \cdots \ell_n \equiv \pm \ell'_1 \cdots \ell'_n k^n \pmod{m}$ for some integer k .
3. Let X be the smash product of k copies of a Moore space $M(\mathbb{Z}_p, n)$ with p prime. Compute the Bockstein homomorphisms in $H^*(X; \mathbb{Z}_p)$ and use this to describe $H^*(X; \mathbb{Z})$.
4. Using the cup product structure in $H^*(SO(5); \mathbb{Z})$, show that $SO(5)$ is not homotopy equivalent to the product of any two CW complexes with nontrivial cohomology.

3.F Limits and Ext

It often happens that one has a CW complex X expressed as a union of an increasing sequence of subcomplexes $X_0 \subset X_1 \subset X_2 \subset \cdots$. For example, X_i could be the i -skeleton of X , or the X_i 's could be finite complexes whose union is X . In situations of this sort, Proposition 3.33 says that $H_n(X; G)$ is the direct limit $\varinjlim H_n(X_i; G)$. Our goal in this section is to show this holds more generally for any homology theory, and to derive the corresponding formula for cohomology theories, which is a bit more complicated even for ordinary cohomology with $\mathbb{Z}$ coefficients. For ordinary homology and cohomology the results apply somewhat more generally than just to CW complexes, since if a space X is the union of an increasing sequence of subspaces X_i with the property that each compact set in X is contained in some X_i , then the singular complex of X is the union of the singular complexes of the X_i 's, and so this gives a reduction to the CW case.

Passing to limits can often result in nonfinitely generated homology and cohomology groups. At the end of this section we describe some of the rather subtle behavior of Ext for nonfinitely generated groups.

Direct and Inverse Limits

As a special case of the general definition in §3.3, the direct limit $\varinjlim G_i$ of a sequence of homomorphisms of abelian groups $G_1 \xrightarrow{\alpha_1} G_2 \xrightarrow{\alpha_2} G_3 \rightarrow \cdots$ is defined to be the quotient of the direct sum $\bigoplus_i G_i$ by the subgroup consisting of elements of the form $(g_1, g_2 - \alpha_1(g_1), g_3 - \alpha_2(g_2), \dots)$. It is easy to see from this definition that every element of $\varinjlim G_i$ is represented by an element $g_i \in G_i$ for some i , and two such representatives $g_i \in G_i$ and $g_j \in G_j$ define the same element of $\varinjlim G_i$ iff they have the same image in some G_k under the appropriate composition of α_ℓ 's. If all the α_i 's are injective and are viewed as inclusions of subgroups, $\varinjlim G_i$ is just $\bigcup_i G_i$.

Example 3F.1. For a prime p , consider the sequence $\mathbb{Z} \xrightarrow{p} \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \cdots$ with all maps multiplication by p . Then $\varinjlim G_i$ can be identified with the subgroup $\mathbb{Z}[1/p]$ of $\mathbb{Q}$ consisting of rational numbers with denominator a power of p . More generally, we can realize any subgroup of $\mathbb{Q}$ as the direct limit of a sequence $\mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \mathbb{Z} \rightarrow \cdots$ with an appropriate choice of maps. For example, if the n^{th} map is multiplication by n , then the direct limit is $\mathbb{Q}$ itself.

Example 3F.2. The sequence of injections $\mathbb{Z}_p \xrightarrow{p} \mathbb{Z}_{p^2} \xrightarrow{p} \mathbb{Z}_{p^3} \rightarrow \cdots$, with p prime, has direct limit a group we denote $\mathbb{Z}_{p^\infty}$. This is isomorphic to $\mathbb{Z}[1/p]/\mathbb{Z}$, the subgroup of $\mathbb{Q}/\mathbb{Z}$ represented by fractions with denominator a power of p . In fact $\mathbb{Q}/\mathbb{Z}$ is isomorphic to the direct sum of the subgroups $\mathbb{Z}[1/p]/\mathbb{Z} \approx \mathbb{Z}_{p^\infty}$ for all primes p . It is not hard to determine all the subgroups of $\mathbb{Q}/\mathbb{Z}$ and see that each one can be realized as a direct limit of finite cyclic groups with injective maps between them. Conversely, every such direct limit is isomorphic to a subgroup of $\mathbb{Q}/\mathbb{Z}$.